

28/03/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

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AIM - 720

CC-318

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(Advanced **INTENSIVE Mastery for 720**)

MM : 720

PST-2

Time : 180 Mins.

- Physics:** Electric Charges and Fields, Electrostatics Potential and Capacitance, Current Electricity, Moving Charges & Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current
- Chemistry:** Solutions, Electrochemistry, Chemical Kinetics, The p-Block Elements(Group-15 to 18), The d & f-Block Elements, Coordination Compounds, Principles Related to Practical Chemistry
- Botany:** Sexual reproduction in flowering plants, Principles of inheritance and variations, Biodiversity and conservation
- Zoology:** Reproductive Health, Human Health & Disease, Evolution

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Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720
720 | **Mridul M Anand**
3 Year Classroom



720
720 | **Ayush Naugraiya**
4 Year Classroom



720
720 | **Arghyadeep Dutta**
2 Year Classroom



720
720 | **Aryan Yadav**
1 Year Classroom



720
720 | **Palansha Agarwal**
2 Year Classroom



720
720 | **Iram Quazi**
1 Year Classroom

PHYSICS

1. A magnet is allowed to fall through a metallic ring. During the fall, acceleration of magnet is

(1) Equal to g
 (2) More than g
 (3) Less than g
 (4) Either (1) or (2)

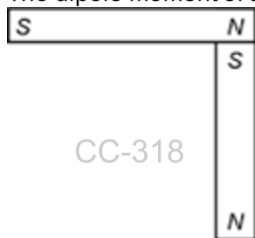
2. Flux (in Wb) in a closed circuit of resistance $5\ \Omega$ varies with time t (in s) according to the equation

$$\phi = 4t^2 - 2t + 1$$

what is the magnitude of the induced current at $t = 1$ s?

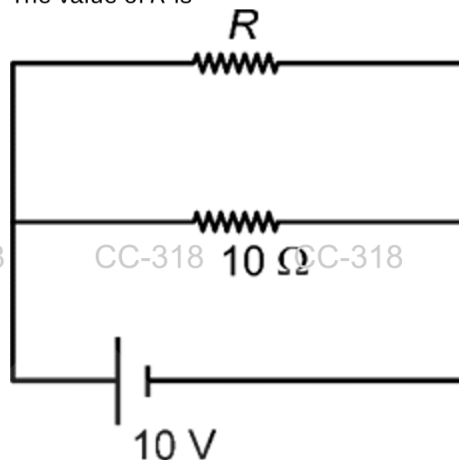
(1) 1 A
 (2) 0.8 A
 (3) 1.4 A
 (4) 1.2 A

3. Two identical bar magnets each of dipole moment P and length l are placed perpendicular to each other as shown. The dipole moment of the combination is



(1) $2P$
 (2) $\sqrt{2}P$
 (3) $\sqrt{3}P$
 (4) $\frac{P}{2}$

4. The power dissipated in the circuit shown in figure is 20 W. The value of R is

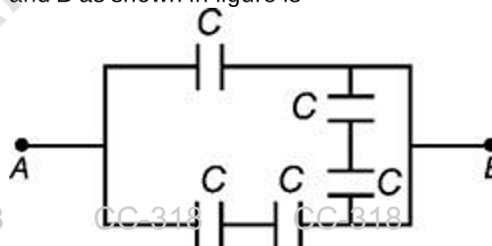


(1) $5\ \Omega$
 (2) $15\ \Omega$
 (3) $10\ \Omega$
 (4) $20\ \Omega$

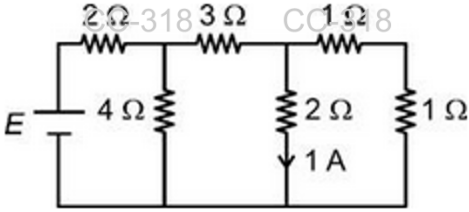
5. The capacitance of an isolated conducting sphere of radius R is proportional to

(1) R^2
 (2) R^{-1}
 (3) R^{-2}
 (4) R

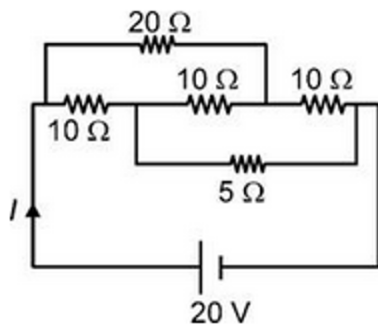
6. The equivalent capacitance of the combination between A and B as shown in figure is



(1) $\frac{C}{3}$
 (2) $\frac{3C}{2}$
 (3) $3C$
 (4) $\frac{C}{2}$

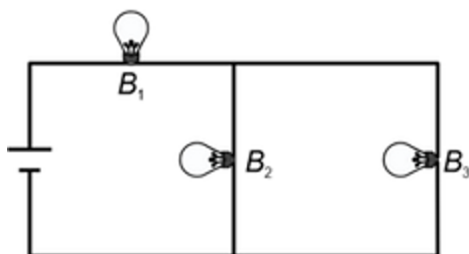
7. A magnetic needle suspended parallel in direction to a magnetic field requires 3 J of work to turn it through 60° . The torque needed to maintain the needle in this position will be
- $\sqrt{3}$ N m
 - $3\sqrt{3}$ N m
 - $6\sqrt{3}$ N m
 - 3 N m
8. A bar magnet has a magnetic moment equal to 25×10^{-4} A m². It is suspended in a magnetic field, which has a magnetic induction B equal to 4×10^{-4} T. The magnet vibrates with a period of vibration equal to 10 s. The moment of inertia of the magnet is [Take $\pi^2 \approx 10$]
- 12×10^{-6} kg m²
 - 2.5×10^{-7} kg m²
 - 1.2×10^{-6} kg m²
 - 25×10^{-7} kg m²
9. A circular loop of area 10 cm², carrying a current 4 A is placed in magnetic field of 2 T perpendicular to the plane of loop. The torque on the loop due to magnetic field is
- 2×10^{-3} N m
 - 4×10^{-3} N m
 - 8×10^{-3} N m
 - Zero
10. A straight wire of mass 50 g and length 1 m carries a current of 4 A. It is suspended in mid-air by a uniform transverse magnetic field B_0 . What is the magnitude of magnetic field?
- 25×10^{-3} T
 - 125×10^{-3} T
 - 50×10^{-3} T
 - 500×10^{-3} T
11. A $\frac{2}{\pi}$ mH inductor is connected to a 100 V, 50 Hz ac source. The inductive reactance of the circuit is
- 0.1 Ω
 - 0.2 Ω
 - 2 Ω
 - 1 Ω
12. A coil of area $\vec{A} = (2\hat{i} - 2\hat{j})$ m² is placed in uniform magnetic field $\vec{B} = (\hat{i} - 2\hat{j} + 4\hat{k})$ T. The flux passing through this coil is
- 2 Wb
 - 10 Wb
 - 8 Wb
 - 6 Wb
13. Electric potential at the surface of a uniformly charged non-conducting solid sphere is 10 V. Find the electric potential at the centre of sphere.
- 5 V
 - 10 V
 - 15 V
 - Zero
14. Electric potential (in V) in a particular region of space is $V = 12x - y^2$ (where x and y are in m). The magnitude of electric field at point P(1 m, -1 m) is
- $12 \frac{V}{m}$
 - $13 \frac{V}{m}$
 - $\sqrt{140} \frac{V}{m}$
 - $\sqrt{148} \frac{V}{m}$
15. Given below are two statements:
Statement I: Gauss's law is true for any closed surface no matter what its shape or size is.
Statement II: Gauss's law is based on the inverse square dependence on distance contained in the Coulomb's law.
 In light of above statements, choose the most appropriate answer from the following.
- Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
 - Both statements I and II are correct
 - Both statements I and II are incorrect
16. The emf of the battery shown in figure is
- 
- 4 V
 - 8 V
 - 12 V
 - 16 V

17. The current I drawn from the battery will be



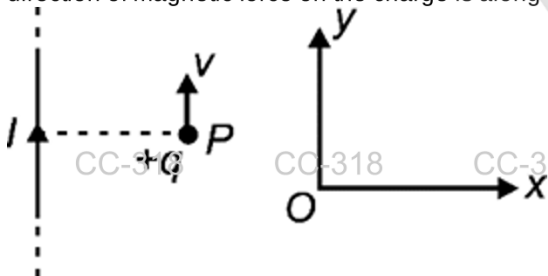
- (1) 1 A
- (2) 0.5 A
- (3) 2 A
- (4) 0.8 A

18. Three identical bulbs B_1 , B_2 and B_3 are connected to the source voltage as shown in figure. If bulb B_2 is fused, then



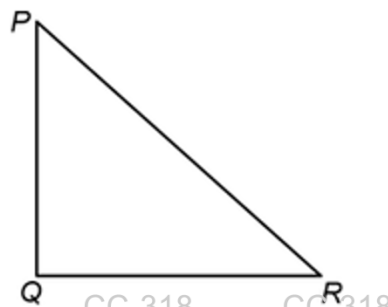
- (1) Brightness of bulb B_1 decreases
- (2) Brightness of bulb B_3 decreases
- (3) Brightness of bulb B_3 increases
- (4) Both (1) and (3)

19. A very long straight wire carries a current I . At the instant when a charge $+q$ at point P has velocity $\vec{v}$ as shown, the direction of magnetic force on the charge is along



- (1) +x-axis
- (2) -x-axis
- (3) +y-axis
- (4) -y-axis

20. A current carrying closed loop in the form of a right angled isosceles triangle PQR is placed in a uniform magnetic field acting along PQ . If the magnetic force on the arm RP is $\vec{F}$, the force on the arm QR is



- (1) $\sqrt{2}\vec{F}$
- (2) $-\sqrt{2}\vec{F}$
- (3) $\vec{F}$
- (4) $-\vec{F}$

21. Given below are two statements, one is the **Assertion (A)** and the other is **Reason (R)**.

Assertion (A) : 80 W, 40 W and 20 W bulbs, each marked 200 V, are connected in series with a voltage source, then 20 W bulb gives maximum illumination.

Reason (R) : In a series circuit, current is same in all the bulbs and resistance of filament of 20 W bulb is maximum.

- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

22. In an AC circuit an alternating voltage $E = 100\sqrt{2} \sin 100t$ V is connected to an inductor of inductance 0.1 H. The rms value of the current in the circuit is

- (1) $10\sqrt{2}$ A
- (2) 10 A
- (3) $5\sqrt{2}$ A
- (4) 20 A

23. Four point charges $q_A = 4 \mu\text{C}$, $q_B = -2 \mu\text{C}$, $q_C = 4 \mu\text{C}$ and $q_D = 2 \mu\text{C}$ are located at the corners of a square ABCD of side $3\sqrt{2}$ m. The net electrostatic force on a 1 C charge placed at the centre of the square will be

- (1) 2 kN
- (2) 4 kN
- (3) 8 kN
- (4) 18 kN

24. Consider a spherical conductor with an irregular shaped cavity inside it with no charge inside the cavity. If this conductor is placed in an external electric field, then which of the following is correct?

(i) Net electric field inside the cavity will be zero.
 (ii) No net charge will be induced on inner surface of the cavity.
 (iii) Electric field lines will be found inside the cavity.
 (iv) Electric field and potential inside the cavity will be zero.

- (1) (i) and (ii) only
 (2) (i), (ii) and (iii) only
 (3) (ii) and (iii) only
 (4) All (i), (ii), (iii) and (iv)

25. Match the Column -I and Column -II.

	Column-I (Magnetic material)		Column-II (Property)
(i)	Diamagnetic	(a)	Feebly attracted by magnet
(ii)	Paramagnetic	(b)	Strongly attracted by magnet
(iii)	Ferromagnetic	(c)	Feebly repelled by magnet

- (1) (i)-(c), (ii)-(b), (iii)-(a)
 (2) (i)-(b), (ii)-(c), (iii)-(a)
 (3) (i)-(a), (ii)-(b), (iii)-(c)
 (4) (i)-(c), (ii)-(a), (iii)-(b)

26. The SI unit of magnetic susceptibility is

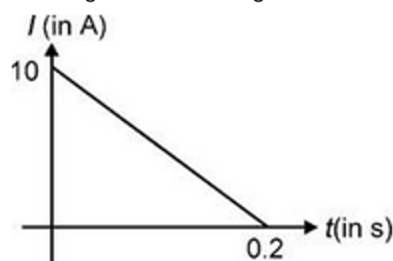
- (1) Henry
 (2) Ampere/metre
 (3) Ampere-metre
 (4) Unitless

27. In a transformer, number of turns in primary and secondary coil are 500 and 1000 respectively. If current in primary coil is 24 A, then current in the secondary coil is

- (1) 24 A
 (2) 12 A
 (3) 48 A
 (4) 6 A

28. In a coil of resistance $5\ \Omega$, the induced current developed by changing magnetic flux through it as a function of time is shown below.

The magnitude of change in flux through the coil (in Wb) is



- (1) 2
 (2) 5
 (3) 10
 (4) 1

29. An alternating voltage $200\sin(1000t)$ V is applied to a series combination of resistance $R = 30\ \Omega$ and an inductor of inductance 0.04 H. The power factor of the circuit is

- (1) 0.5
 (2) 0.3
 (3) 1.33
 (4) 0.6

30. Given below are two statements, one is the **Assertion (A)** and the other is **Reason (R)**.

Assertion A: In electrostatics, electric field lines are always perpendicular to surface of charged metallic conductor.

Reason R: In electrostatics, surface of charged metallic conductor is always equipotential surface.

- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false

31. RMS value of an alternating voltage given by $E = E_1\sin\omega t + E_2\cos\omega t$, is

- (1) $\sqrt{E_1^2 + E_2^2}$
 (2) $\frac{E_1}{\sqrt{2}}$
 (3) $\frac{E_2}{\sqrt{2}}$
 (4) $\sqrt{\frac{E_1^2 + E_2^2}{2}}$

32. A direct current of 2 A and an alternating current of peak value 2 A pass through two identical resistances separately for same time t . Ratio of heat produced in the two resistances is

(1) 2 : 1
 (2) 4 : 1
 (3) 1 : $\sqrt{2}$
 (4) 1 : 1

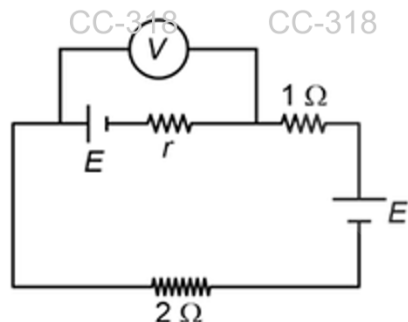
33. If a glass rod is rubbed with silk, it acquires a positive charge because

(1) Protons are added to it
 (2) Protons are removed from it
 (3) Electrons are added to it
 (4) Electrons are removed from it

34. Two balls carrying charge $+2 \mu\text{C}$ and $-2 \mu\text{C}$ attract each other with a force F . If $+2 \mu\text{C}$ is added to both separately, the force between them will be

(1) Zero
 (2) F
 (3) $4F$
 (4) $2F$

35. In the given circuit the reading of ideal voltmeter connected across a battery is $\frac{E}{4}$, then internal resistance r of battery is

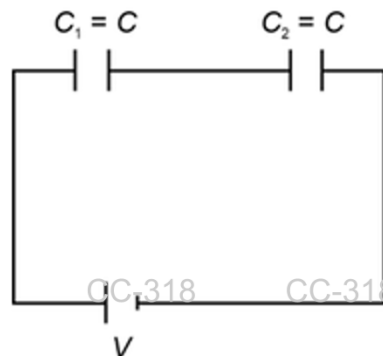


(1) $\frac{1}{7} \Omega$
 (2) $\frac{3}{7} \Omega$
 (3) $\frac{9}{5} \Omega$
 (4) $\frac{5}{9} \Omega$

36. An electron is projected with some velocity along the axis of a current carrying long solenoid. Which of the following is true?

(1) The electron will be accelerated along the axis
 (2) The path of electron will be circular about the axis
 (3) The path of electron will be helical
 (4) The electron will continue to move with same velocity along the axis of solenoid

37. Two identical capacitors of capacitance C are connected in series with a battery. They are fully charged. Now a dielectric slab is inserted between the plates of capacitor (C_1). The potential difference across other capacitor (C_2) will

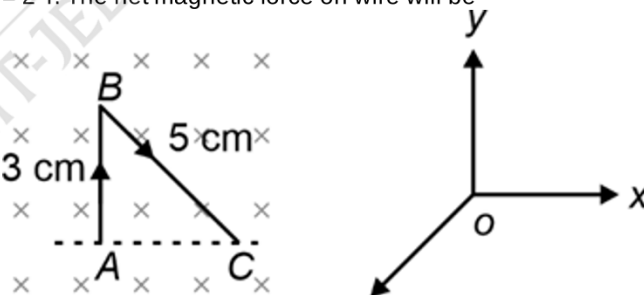


(1) Increase
 (2) Decrease
 (3) Remain same
 (4) Depend on internal resistance of cell

38. If number of turns in a moving coil galvanometer is doubled keeping other parameters constant, then the current sensitivity of galvanometer will be

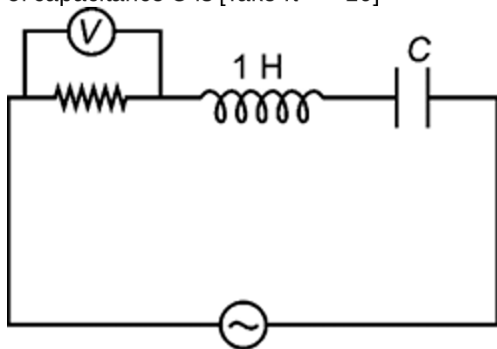
(1) Halved
 (2) Doubled
 (3) Four times
 (4) Same as before

39. A current of 2 A flows through a conducting wire ABC as shown below. The wire is kept in a uniform magnetic field $B = 2 \text{ T}$. The net magnetic force on wire will be



(1) 0.08 N along y-axis
 (2) 0.16 N along y-axis
 (3) 0.08 N along x-axis
 (4) 0.08 N along z-axis

40. In the circuit shown below, the voltmeter reads 100 V. Value of capacitance C is [Take $\pi^2 \approx 10$]



- (1) $5 \mu\text{F}$
 (2) $2.5 \mu\text{F}$
 (3) $20 \mu\text{F}$
 (4) $10 \mu\text{F}$
41. A long straight wire carries a current of 8 A. What is the magnitude of the magnetic field at a point 2 cm from the wire?
- (1) 8 T
 (2) 80 T
 (3) 8 μT
 (4) 80 μT
42. A regular hexagon has side of 0.9 m and has charge of $4 \mu\text{C}$ at each of its vertices. How much work is required in slowly moving a charge of $2 \mu\text{C}$ from infinity to centre of the hexagon?
- (1) 0.12 J
 (2) 0.48 J
 (3) 1.2 J
 (4) 4.8 J

43. A point charge $8.85 \mu\text{C}$ is at a distance 25 cm directly above the centre of a square of side length 50 cm. What is the magnitude of electric flux through the square?

- (1) $1.67 \times 10^4 \text{ V m}$
 (2) $1.67 \times 10^5 \text{ V m}$
 (3) $1.67 \times 10^6 \text{ V m}$
 (4) 10^6 V m

44. A $200 \mu\text{F}$ capacitor is charged by a 100 V supply. It is then disconnected from the supply and is connected to an uncharged $400 \mu\text{F}$ capacitor. The final potential difference across $400 \mu\text{F}$ capacitor will be

- (1) 100 V
 (2) $\frac{100}{3} \text{ V}$
 (3) $\frac{200}{3} \text{ V}$
 (4) 50 V

45. 20 identical cells each of emf E and internal resistance r are connected in a similar manner in series combination. If terminals of any 2 cells is reversed, then equivalent emf of circuit is

- (1) $20E$
 (2) $18E$
 (3) $16E$
 (4) $12E$

CHEMISTRY

46. In two molal solution which contains 0.25 mol solute, there is
- (1) 125 mL solution
 (2) 125 g solvent
 (3) 250 mL solution
 (4) 250 g solvent

47. Given below are two statements.

Statement I: For aqueous solution with density of 1 g cm^{-3} , 1 molal NaCl solution will be more concentrated than 1 molar NaCl solution.

Statement II: Mole fraction of solute is temperature independent.

In the light of above statements, choose the most appropriate option from the options given below.

- (1) Both statement I and statement II are incorrect
 (2) Both statement I and statement II are correct
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

48. Match gases, in list-I with their values of Henry's law constant K_H in kbar at 298 K in list-II.

	List-I		List-II
a.	Ar	(i)	1.83×10^{-5}
b.	HCHO	(ii)	1.67
c.	CO ₂	(iii)	40.3
d.	CH ₂ = CHCl	(iv)	0.611

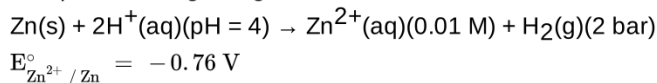
Correct match is

- (1) a(iii), b(i), c(ii), d(iv)
 (2) a(iv), b(ii), c(i), d(iii)
 (3) a(iii), b(ii), c(i), d(iv)
 (4) a(iv), b(iii), c(ii), d(i)
49. Vapour pressure of CHCl₃ and CH₂Cl₂ at 298 K are 200 mmHg and 415 mmHg respectively, then mole fraction of CHCl₃ in vapour phase if 51 g of CHCl₃ is mixed with 80 g CH₂Cl₂ at 298 K will be
- (1) 0.36
 (2) 0.82
 (3) 0.64
 (4) 0.18
50. Which among the following solution will have highest freezing point (electrolytes are 100% ionised).
- (1) 0.1 m NaCl (aq)
 (2) 0.3 m Urea (aq)
 (3) 0.5 m C₆H₁₂O₆ (aq)
 (4) 0.2 m CaCl₂ (aq)
51. Given below are two statements.
Statement I: 0.1 m NaCl aqueous solution has higher boiling point than 0.1 m KCl aqueous solution if both are completely ionised.
Statement II: Molal elevation constant for NaCl aqueous solution is higher than KCl aqueous solution.
 In the light of above statements, choose the most appropriate option from the options given below.
- (1) Both statement I and statement II are incorrect
 (2) Statement I is correct but statement II is incorrect
 (3) Both statement I and statement II are correct
 (4) Statement I is incorrect but statement II is correct
52. 0.1 M CaCl₂ with 80% ionisation will be isotonic with
- (1) 0.13 M KCl aqueous solution with 100% ionisation
 (2) 0.3 M urea aqueous solution
 (3) 0.1 M Na₂SO₄ aqueous solution with 90% ionisation
 (4) 0.2 M sucrose aqueous solution

53. Correct expression for spontaneous cell is

- (1) $\Delta G < 0, E_{\text{cell}} < 0$
 (2) $\Delta G > 0, E_{\text{cell}} > 0$
 (3) $\Delta G < 0, E_{\text{cell}} > 0$
 (4) $\Delta G > 0, E_{\text{cell}} < 0$

54. Cell potential of given galvanic cell at 298 K will be



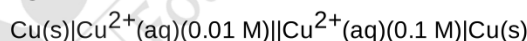
- (1) 0.84 V
 (2) 0.5 V
 (3) 0.58 V
 (4) 0.92 V

55. Consider the following statements

- (a) Ag is more reactive metal than Cu
 (b) Al³⁺/Al has more negative standard reduction potential than Zn²⁺/Zn.
 (c) Cl₂ + Br⁻(aq) → Br₂ + Cl⁻(aq) is a spontaneous reaction
 Correct statement(s) is/are

- (1) (a) and (b) only
 (2) (b) and (c) only
 (3) (a) only
 (4) (a), (b) and (c)

56. K_c for given cell representation will be



- (1) 10²
 (2) 10⁻²
 (3) 10
 (4) 1

57. Given below are two statements, one is the assertion (A) and the other is reason (R).

Assertion (A): Specific conductance (k) decreases with dilution.

Reason (R): On dilution number of ions per ml of solution decrease.

In the light of above statements, select the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true

58. Match commercial cell in list-I with their anode in list-II.

	List-I		List-II
a.	Leclanche cell	(i)	Zn-Hg amalgam
b.	Mercury cell	(ii)	Cd
c.	Lead storage Battery	(iii)	Zn
d.	Nickel-Cadmium cell	(iv)	Pb

Correct match is

- (1) a(iii), b(iv), c(i), d(ii)
 (2) a(iv), b(ii), c(iii), d(i)
 (3) a(iii), b(i), c(iv), d(ii)
 (4) a(ii), b(i), c(iv), d(iii)

59. Amount of charge required to produce 1 mol $O_2(g)$ from oxidation of water will be

- (1) 1 F
 (2) 2 F
 (3) 3 F
 (4) 4 F

60. For a certain chemical reaction the, rate = $k[A]^2[B]^{-1}$ if concentration of both A and B are doubled then the rate would be

- (1) Increase with a factor of 8
 (2) Increase with a factor of 6
 (3) Increase with a factor of 2
 (4) Increase with a factor of 4

61. The experimental data for a reaction $2X + Y \rightarrow X_2Y$ is

Exp.	[X] M	[Y] M	Rate (Ms^{-1})
1	0.1	0.2	1.2×10^{-3}
2	0.1	0.1	3×10^{-4}
3	0.2	0.2	2.4×10^{-3}

The rate law expression for above data is

- (1) $r = k[X]^2[Y]$
 (2) $r = k[X][Y]^2$
 (3) $r = k[X][Y]$
 (4) $r = k[X]^2[Y]^2$

62. For a chemical reaction starting with 2 M reactant and rate constant $2.5 \times 10^{-2} Ms^{-1}$, time of completion of reaction will be

- (1) 40 s
 (2) Infinite
 (3) 80 s
 (4) 160 s

63. For a first order reaction, rate of reaction at 20 s is $0.001 mol L^{-1} s^{-1}$ while concentration at this moment is 0.1 M. Half life of the reaction will be

- (1) 69.3 s
 (2) 40 s
 (3) 34.51 s
 (4) 80 s

64. If half life of a first order reaction is 10 min then $t_{99.9\%}$ of reaction will be nearly

- (1) 70 min
 (2) 50 min
 (3) 80 min
 (4) 100 min

65. Given below are two statements.

Statement I: Hydrolysis of ethyl acetate in acidic medium is a pseudo first order reaction.

Statement II: Decomposition of gaseous NH_3 on a hot platinum surface at high pressure is a first order reaction. In the light of above statements, choose the most appropriate option from the options given below.

- (1) Both statement I and statement II are incorrect
 (2) Statement I is correct but statement II is incorrect
 (3) Both statement I and statement II are correct
 (4) Statement I is incorrect but statement II is correct

66. Given below are two statements.

Statement I: Rate constant increases with increase in temperature for endothermic reaction.

Statement II: Rate constant decreases with increase in temperature for exothermic reaction. In the light of above statements, choose the most appropriate option from the options given below.

- (1) Both statement I and statement II are incorrect
 (2) Statement I is correct but statement II is incorrect
 (3) Both statement I and statement II are correct
 (4) Statement I is incorrect but statement II is correct

67. Match oxides in list-I with their nature in list-II.

	List-I		List-II
a.	N_2O	(i)	Basic
b.	N_2O_3	(ii)	Neutral
c.	As_2O_3	(iii)	Amphoteric
d.	Bi_2O_3	(iv)	Acidic

Correct match is

- (1) a(ii), b(iv), c(iii), d(i)
 (2) a(iv), b(iii), c(ii), d(i)
 (3) a(iii), b(i), c(iv), d(ii)
 (4) a(iv), b(iii), c(i), d(ii)

68. Basicity of H_3PO_2 is

- (1) 2
- (2) 3
- (3) 0
- (4) 1

69. Correct order of boiling point of group 16 hydrides is

- (1) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$
- (2) $\text{H}_2\text{Te} < \text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{O}$
- (3) $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{O}$
- (4) $\text{H}_2\text{S} < \text{H}_2\text{O} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$

70. Which among the following elements has least negative electron gain enthalpy?

- (1) S
- (2) O
- (3) Se
- (4) Te

71. Correct order of bond dissociation enthalpy of halogen molecules is

- (1) $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$
- (2) $\text{I}_2 < \text{Br}_2 < \text{Cl}_2 < \text{F}_2$
- (3) $\text{I}_2 < \text{Cl}_2 < \text{Br}_2 < \text{F}_2$
- (4) $\text{I}_2 < \text{F}_2 < \text{Br}_2 < \text{Cl}_2$

72. Incorrect statement among the following is

- (1) HOBrO is least likely to exist
- (2) Shape of ClF_3 is bent T-shape
- (3) HClO_4 is less acidic than HClO
- (4) Chlorine reacts with hot and conc. NaOH to produce sodium chlorate

73. Correct hybridisation and shape for XeF_6 respectively are

- (1) sp^3d^2 and octahedral
- (2) sp^3d^3 and pentagonal bipyramidal
- (3) sp^3d^2 and distorted octahedral
- (4) sp^3d^3 and distorted octahedral

74. Match species in list-I with their spin only magnetic moment in list-II.

	List-I		List-II
a.	Sc^{3+}	(i)	$\sqrt{8}$ BM
b.	Ni^{2+}	(ii)	0 BM
c.	Fe^{3+}	(iii)	$\sqrt{15}$ BM
d.	Co^{2+}	(iv)	$\sqrt{35}$ BM

Correct match is

- (1) a(ii), b(i), c(iv), d(iii)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(ii), b(iv), c(i), d(iii)
- (4) a(iii), b(i), c(iv), d(ii)

75. Hybridisation and shape of $[\text{Sc}(\text{H}_2\text{O})_6]\text{Cl}_3$ respectively are

- (1) dsp^2 and square planar
- (2) sp^3d^2 and octahedral
- (3) sp^3 and tetrahedral
- (4) d^2sp^3 and octahedral

76. Given below are two statements, one is the assertion (A) and the other is reason (R).

Assertion (A): Aqueous solution of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ exhibits colour.

Reason (R): In $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, d-d transition of electron is possible.

In the light of above statements, select the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

77. On reaction of KMnO_4 with I^- in the acidic medium oxidation state of Mn changes from

- (1) +7 to +2
- (2) +7 to +4
- (3) +7 to +6
- (4) +7 to 0

78. Fusion of chromite ore with sodium carbonate in free access of air results in the formation of

- (1) $\text{Na}_2\text{Cr}_2\text{O}_7$, FeO and CO_2
- (2) $\text{Na}_2\text{Cr}_2\text{O}_7$, Fe_2O_3 and CO_2
- (3) Na_2CrO_4 , Fe_2O_3 and CO_2
- (4) Na_2CrO_4 , Fe_3O_4 and CO

79. Observed change in colour of $K_2Cr_2O_7$ if it is exposed to basic medium is

- (1) Orange to green
- (2) Green to orange
- (3) Yellow to orange
- (4) Orange to yellow

80. Which among the following can exhibit +4 oxidation state?

- (1) Pr
- (2) Eu
- (3) Yb
- (4) Sm

81. Identify the **incorrect** match.

(1)	Hf	—	Transition element
(2)	No	—	Lanthanoid
(3)	Md	—	Actinoid
(4)	Tm	—	Lanthanoid

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

82. Which among the following can show optical isomerism?

- (1) $cis-[PtCl_2(NH_3)_2]$
- (2) $[Co(ox)_3]^{3-}$
- (3) $[Fe(CN)_6]^{4-}$
- (4) $[Pt(en)_2]^{2+}$

83. Which among the following metal carbonyls has strongest metal carbon bond?

- (1) $[Fe(CO)_5]$
- (2) $[V(CO)_6]^-$
- (3) $[Cr(CO)_6]$
- (4) $[Mn(CO)_6]^+$

84. The correct order of increasing field strength of ligands to form co-ordination compounds is

- (1) $NH_3 < en < CN^\ominus < edta^{4-}$
- (2) $CN^\ominus < en < NH_3 < edta^{4-}$
- (3) $CN^\ominus < edta^{4-} < en < NH_3$
- (4) $edta^{4-} < NH_3 < en < CN^\ominus$

85. Correct electronic configuration of the central atom in $[Co(H_2O)_6]Cl_3$ based on crystal field theory is

- (1) $t_{2g}^6 e_g^0$
- (2) $t_{2g}^4 e_g^2$
- (3) $t_{2g}^5 e_g^2$
- (4) $t_{2g}^3 e_g^3$

86. Match group of cation in list-I with their group reagent of qualitative analysis in list-II.

	List-I		List-II
a.	Group-I	(i)	$NH_4OH + NH_4Cl$
b.	Group-II	(ii)	Dilute HCl
c.	Group-III	(iii)	$H_2S + \text{dilute HCl}$
d.	Group-IV	(iv)	$H_2S + NH_4OH$

Correct match among the following is

- (1) a(iii), b(ii), c(iv), d(i)
- (2) a(iii), b(ii), c(i), d(iv)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(ii), b(iii), c(i), d(iv)

87. Heat evolved on neutralisation of 1 mol NaOH with 1 mol HCl in aqueous solution is nearly

- (1) 57.1 kJ
- (2) 13.7 kJ
- (3) 44.8 kJ
- (4) 36.5 kJ

88. Given below are two statements.

Statement I: Carbonate salts gives brisk effervescence on reaction with dilute H_2SO_4 .

Statement II: Cl^- ion can be detected by dilute H_2SO_4 .

In the light of above statements, choose the most appropriate option from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

89. Lyophilic and lyophobic colloids among the following respectively are

- (1) Egg albumin and arsenic sulphide
- (2) Starch and egg albumin
- (3) Arsenic sulphide and ferric hydroxide
- (4) Aluminium hydroxide and starch

90. Match indicator in list-I with their working pH range in list-II.

	List-I		List-II
a.	Methyl orange	(i)	4.2 to 6.2
b.	Methyl red	(ii)	6.2 to 8.2
c.	Phenol red	(iii)	8.2 to 10.2
d.	Phenolphthalein	(iv)	3.1 to 4.5

Correct match is

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(ii), b(iii), c(iv), d(i)
- (3) a(iv), b(i), c(ii), d(iii)
- (4) a(iii), b(ii), c(i), d(iv)

BOTANY

91. In angiosperms, male gametophyte has an outer wall layer called X, made up of Y.

In the light of above statement, **correctly** fill in the blank for 'X' and 'Y'.

- (1) X → Exothecium, Y → Peptidoglycan
- (2) X → Sporoderm, Y → Cellulose and Chitin
- (3) X → Exine, Y → Sporopollenin
- (4) X → Intine, Y → Pectin

92. Which of the following characters of pea plant expresses itself in both homozygous and heterozygous condition?

- (1) Yellow pod colour
- (2) Terminal position of flower
- (3) Inflated pod shape
- (4) Wrinkled seed shape

93. Read the given statements and choose the **correct** option.

Statement A: An individual affected with Klinefelter's syndrome has overall masculine development with gynaecomastia. It is caused due to the presence of an additional copy of X-chromosome.

Statement B: Human skin colour is an example of inheritance, in which multiple genes are involved, with each allele showing an additive effect and also exhibit the influence of environment.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statement A and statement B are correct
- (4) Both statement A and statement B are incorrect

94. In a typical monocot embryo, a hollow foliar structure which encloses shoot apex and few leaf primordia is called

- (1) Coleorhiza
- (2) Epiblast
- (3) Scutellum
- (4) Coleoptile

95. Invariably autogamous flowers, which are characterized by producing assured seed set, are found in

- (1) *Ficus* and Grasses
- (2) Common pansy and *Oxalis*
- (3) *Zostera* and *Viola*
- (4) *Yucca* and *Salvia*

96. The term used for the type of pollination which is functionally a cross-pollination and genetically similar to autogamy is

- (1) Chasmogamy
- (2) Geitonogamy
- (3) Cleistogamy
- (4) Xenogamy

97. Read the given assertion (A) and reason (R) and choose the **correct** option.

Assertion (A): Sickle cell anaemia is a quantitative problem of synthesising too few globin molecules.

Reason (R): It is caused due to single base substitution at sixth codon of beta globin gene present on chromosome 11.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is false but (R) is true
- (4) Both (A) and (R) are false

98. Select the **correctly** matched pair.

(1)	Sacred groves	–	Chanda and Bastar areas of Meghalaya
(2)	Insects	–	They are the least species-rich taxonomic group among animals
(3)	Genetic diversity	–	Maximum in rice as compared to mango in India
(4)	Biosphere Reserve	–	Buffer zone comprises an undisturbed and legally protected ecosystem

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

99. The recombination frequency between the genes X and T is 56%, A and T is 9%, X and Y is 8% and X and A is 47%. What will be the sequence of these genes on a linear chromosome?

- (1) Y A T X
- (2) X Y T A
- (3) X Y A T
- (4) X A Y T

100. In Honey bees,

- (1) Drones are haploid
- (2) Queen develops by parthenogenesis
- (3) Males do not have a grandfather and grandsons
- (4) Queens produce gametes by mitosis

101. According to Rivet popper hypothesis, rivets on the wings represent

- (1) An ecosystem
- (2) The key species
- (3) Genetic diversity
- (4) An alien species

102. Gene for which of the given feature/disorders can exhibit multiple phenotypic expressions?

- (1) Haemophilia
- (2) Colour blindness
- (3) Phenylketonuria
- (4) Human height

103. Long, ribbon-like pollen grains, generally surrounded by a mucilaginous covering are seen in

- (1) *Eichornia*
- (2) *Zostera*
- (3) *Amorphophallus*
- (4) Water lily

104. Select the **mismatched** pair.

(1)	Phenylketonuria	–	Autosomal recessive trait
(2)	α Thalassemia	–	Controlled by HBB gene on chromosome 11
(3)	T.H. Morgan	–	Coined the term recombination
(4)	Law of segregation	–	Factors or alleles of a pair segregate from each other during gamete formation

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

105. Among the below given options, the highest number of species of birds are found to be present in

- (1) New York
- (2) Greenland
- (3) India
- (4) Colombia

106. In which of the following options, the female has a pair of dissimilar sex chromosomes 'ZW'?

- (1) Grasshopper
- (2) *Drosophila*
- (3) *Homo sapiens*
- (4) Birds

107. How many different kinds of gametes will be produced by pea plants, having the genotypes AabbCcDD and aaBbCCDD respectively?

- (1) Nine and two
- (2) Eight and four
- (3) Four and four
- (4) Two and eight

108. Select the **odd** one out w.r.t. the Law of Dominance.

- (1) Factors occur in pairs
- (2) Characters are controlled by factors
- (3) Dominance is an autonomous feature of a gene or the product that it has information for
- (4) It is used to explain the expression of only one of the parental characters in monohybrid cross in F_1

109. The residual, persistent diploid nutritive tissue found in the seeds of black pepper is known as

- (1) Sporogenous tissue
- (2) Perisperm
- (3) Integument
- (4) Endosperm

110. State True (T) or False (F) for the given statements.

- (A) World Summit on Sustainable Development was held in 1992 in Johannesburg, South America.
- (B) Biodiversity loss in a region may lead to decreased variability in certain ecosystem processes such as pest and disease cycles.
- (C) Ecologists have discovered that the value of Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or region, eg. plants in Britain or birds in California.
- Choose the **correct** option.

	(A)	(B)	(C)
(1)	T	T	F
(2)	F	F	T
(3)	T	T	T
(4)	F	T	F

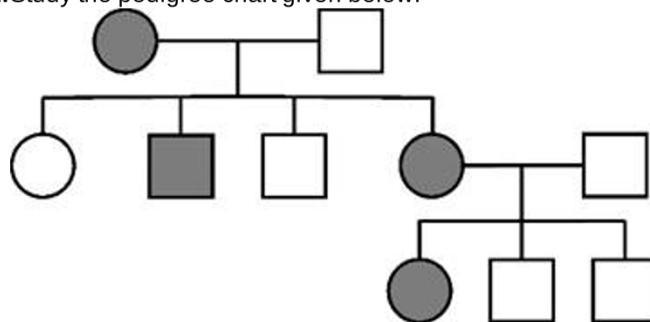
- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

111. Select the **correct** option with respect to apomixis.

- (i) Seeds developed through apomixis, have advantages in horticulture and agriculture.
- (ii) It is a form of asexual reproduction that mimics sexual reproduction.
- (iii) It is seen in plants such as some species of Asteraceae and grasses.
- (iv) Apomictic seeds show segregation of characters in the hybrid progeny.

- (1) Only (i) and (iv) are correct
- (2) Only (ii) and (iii) are correct
- (3) Only (i), (ii) and (iii) are correct
- (4) All (i), (ii), (iii) and (iv) are correct

112. Study the pedigree chart given below:



Which of the following is **not** shown by the above pedigree?

- (1) Inheritance of a sex-linked recessive disease
- (2) The genotype of all the unaffected males can be Aa or aa
- (3) Inheritance of an autosomal dominant trait like myotonic dystrophy
- (4) All affected females can be homozygous or heterozygous

113. Read the given statements.

Statement A: In a typical test cross, an organism showing a dominant phenotype is crossed with the recessive parent instead of self-crossing.

Statement B: The number of contrasting characters studied by Mendel for his experiments was 14.

Choose the **correct** option.

- (1) Only statement B is correct
- (2) Only statement A is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

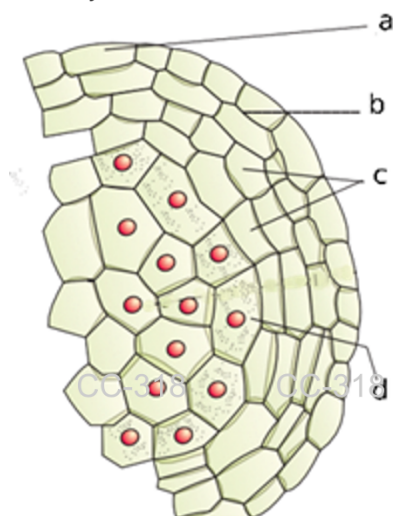
114. A man is suffering from haemophilia which occurs due to the presence of a defective form of blood clotting protein. This trait bearing chromosome can be inherited from this man to his

- (1) 50% daughters only
- (2) Sons only
- (3) Both daughter and son
- (4) Both daughter and grandson

115. Which of the following statements is **not** correct?

- (1) Emasculation is needed if the female parent produces unisexual flowers
- (2) Gecko lizard and garden lizard have also been reported as pollinators in some species
- (3) In angiosperms, during female gametophyte formation, after degeneration of 3 megaspores, 3 successive free-nuclear divisions occur in the functional megaspore
- (4) Each cell of sporogenous tissue is capable of giving rise to a microspore tetrad

116. Examine the figure given below and select the option that correctly identifies and describes the parts a, b, c and d.



- (1) 'a' represents epidermis which helps in anther dehiscence due to its hygroscopic nature
- (2) 'b' represents a wall layer which nourishes the developing pollen grains
- (3) 'c' represents a wall layer which surrounds the sporogenous tissue at maturity
- (4) 'd' represents a wall layer, containing cells with dense cytoplasm and are polyploid

117. Select the **odd** one out w.r.t number of megasporangia present in an ovary.

- (1) Wheat
- (2) Orchids
- (3) Paddy
- (4) Mango

118. If a male with blood group AB marries a female with blood group B, then which of the following observations will **not** be true about the given situation?

- (1) 50% of the progeny obtained can have 'B' blood group
- (2) 25% of the progeny obtained can have AB blood group
- (3) In progeny, 'A' blood group cannot appear in case of parents having $I^A I^B \times I^B I^B$ genotype
- (4) 25% of the progeny obtained can have 'O' blood group

119. A special cellular thickening at micropylar tip, which plays an important role in guiding the pollen tube into synergid is

- (1) Co-operative cells
- (2) Vegetative cells
- (3) Filiform apparatus
- (4) Egg apparatus

120. All of the following are **true** for endosperm, **except**

- (1) Its development precedes embryo development.
- (2) The coconut water from tender coconut is free-nuclear endosperm.
- (3) In castor, it is completely consumed before seed maturation.
- (4) It is a product of triple fusion.

121. Match List-I with List-II.

	List-I		List-II
(a)	Mutagens	(i)	Deletions and insertions of base pairs of DNA
(b)	Polyploidy	(ii)	Arises due to non-disjunction of two chromosomes of homologous pair
(c)	Aneuploidy	(iii)	It is often seen in plants, resulted from failure of cytokinesis after telophase
(d)	Frame-shift mutation	(iv)	Agents that artificially produce mutations like UV rays

Choose the **correct** option.

- (1) a(iii), b(i), c(ii), d(iv)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(i), b(iv), c(iii), d(ii)
- (4) a(iv), b(i), c(iii), d(ii)

122. Threatened species of plants and animals are taken out from their natural habitat and placed in special setting to protect them and given special care in

- (1) Zoological parks and botanical gardens
- (2) National parks and wildlife sanctuaries
- (3) Sacred groves and wildlife safari parks
- (4) Biosphere reserves and botanical gardens

123. How many of the given features are **true** for maize and sugarcane?

- (a) Absence of nectaries
 - (b) Presence of usually single ovule in ovary
 - (c) Flower packed into inflorescence
 - (d) Sticky and light pollen grains
 - (e) Colourful fragrant flowers
- Choose the **correct** option.

- (1) Two
- (2) Three
- (3) Four
- (4) Five

124. "Humans derive countless direct economic benefits from nature", this is exemplified by all of the following, **except**

- (1) Firewood
- (2) Construction material
- (3) Industrial products
- (4) Aesthetic pleasure

125. Select the **odd** one out w.r.t. a hotspot of biodiversity in India.

- (1) Indo-Burma
- (2) Himalaya
- (3) Aravalli hills
- (4) Western ghats

126. In comparison to temperate areas, tropics are

- (1) less productive
- (2) less seasonal
- (3) less constant and predictable
- (4) less species diverse

127. Select the option with **correct** set of statements (A-D)

- (A) The F_2 phenotypic ratio obtained in a Mendelian dihybrid cross and incomplete dominance will be 9 : 3 : 3 : 1 and 1 : 2 : 1, respectively.
- (B) Loosely linked genes present on the same chromosome show similar recombination percentage as the tightly linked ones.
- (C) The phenotype/trait is dependent on the functioning of the allele.
- (D) Mendel's work remained unrecognized till 1900 due to one of the reason that his approach of using mathematics to explain biological phenomena was new and unacceptable.

- (1) All except (B)
- (2) (A) and (D) only
- (3) (B) and (D) only
- (4) (A) and (C) only

128. Which of the following taxa has maximum number of species in Amazonian rain forest in South America?

- (1) Fishes
- (2) Amphibians
- (3) Reptiles
- (4) Birds

129. Match List-I with List-II.

List-I	List-II
(a) Self-incompatibility	(i) Spindle shaped with dense cytoplasm and a nucleus
(b) <i>Michelia</i>	(ii) Prevents autogamy as well as geitonogamy
(c) Generative cell	(iii) Genetically controlled mechanism to prevent self-pollination
(d) Dioecious condition	(iv) Apocarpous gynoecium

Choose the **correct** option.

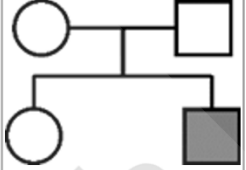
- (1) a(iii), b(i), c(iv), d(ii)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(iii), b(iv), c(i), d(ii)
- (4) a(i), b(iii), c(ii), d(iv)

130. Select the **correctly** matched pair.

(1) Epicotyl	—	The cylindrical portion of embryonal axis below the level of cotyledons.
(2) <i>Phoenix dactylifera</i>	—	Oldest viable seed excavated from Arctic Tundra after 10,000 years of dormancy.
(3) Pollen viability	—	Maintained for months in rice and wheat.
(4) Corn cob	—	Tassels represents stigma and style

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

131. Select the **incorrectly** matched pair w.r.t symbols used in human pedigree analysis.

(1)		Parents with affected female child
(2)		Affected individuals
(3)		Sex unspecified
(4)		Consanguineous mating

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

132. In garden pea, the gene controlling starch synthesis has two alleles 'B' and 'b'. Which of the following characteristics of inheritance can be observed if two plants with Bb genotype were hybridized?

- (a) Incomplete dominance
 - (b) Pleiotropy
 - (c) Dominance
 - (d) Codominance
 - (e) 75% of the total seeds are round and contain large sized starch grains
- Choose the **correct** ones.

- (1) (c), (d) and (e) only
- (2) (c), (e) only
- (3) Only (a), (b) and (c)
- (4) All are correct

133. Extinction of A and B in the last 500 years, were due to overexploitation by humans.
Select the option which correctly fill in the blanks for A and B.

- (1) A → Dodo, B → Cichlid fish
- (2) A → Thylacine, B → Nile perch
- (3) A → Passenger pigeon, B → Steller's sea cow
- (4) A → *Clarias gariepinus*, B → *Lantana*

134. In humans, ABO blood grouping does not show all given pattern of inheritance, **except**

- (1) Multiple allelism
- (2) Incomplete dominance
- (3) Polygenic inheritance
- (4) Pleiotropy

135. Select the **odd** one out with respect to Down's syndrome.

- (1) Small round head
- (2) Small and wrinkled tongue
- (3) Mental development is retarded
- (4) Trisomy of 21 chromosome

ZOOLOGY

136. The genetic material of HIV is

- (1) Two single stranded RNA
- (2) One single stranded RNA
- (3) Double stranded RNA
- (4) Double stranded DNA

137. Select the correct set of drugs which are extracted from the same plant.

- (1) Morphine, Heroin, Coke
- (2) Hashish, Charas, Morphine
- (3) Crack, Cocaine, Marijuana
- (4) Ganja, Hashish, Marijuana

138. All of the following are true w.r.t. the ill-effects of smoking on human body, **except**

- (1) Increased chances of developing gastric ulcers, bronchitis and coronary artery disease.
- (2) Decreased CO content in blood as affinity of CO is less for Hb than O₂
- (3) Increased risk of cancer of lungs, urinary bladder and throat.
- (4) Oxygen deficiency in the body

139. A patient with complaint of abdominal pain visits a doctor. The doctor also observes:

- (a) Internal bleeding
- (b) Anemia
- (c) Blockage of intestinal passage

Based on the above mentioned observations the doctor concluded that patient is most likely suffering from

- (1) Amoebic dysentery
- (2) Ringworm
- (3) Mumps
- (4) Ascariasis

140. Choose the correct statement.

- (1) Introducing *Gambusia* in ponds or nearby water bodies is not helpful in controlling the spread of chikungunya.
- (2) Diseases like tetanus, polio and diphtheria have been controlled to large extent by the use of vaccines.
- (3) The principle of ELISA is based upon the memory of the immune system.
- (4) The Indian population doubled from the year of independence to the year 2011.

141. A French naturalist Lamarck postulated that

- (1) Evolution is driven by the use and disuse of organs
- (2) Reproductive fitness is the ultimate fitness
- (3) Inheritable 'factors' influence phenotype
- (4) Mutations are random and directionless

142. The contraceptives which require help of a doctor or expert nurses for their insertion in the body of a human female include

- (1) Vaults and condoms
- (2) Lippes loop and LNG-20
- (3) Diaphragms and nirodh
- (4) CuT and spermicidal jellies

143. First mammals were

- (a) Like shrews
 - (b) Small sized
 - (c) Intelligent in sensing and avoiding danger
- Select the **correct** option.

- (1) Only (a) and (b)
- (2) (a), (b) and (c)
- (3) Only (c) and (b)
- (4) Only (a)

144. Comprehend the given statements

Statement A: Spermicidal creams, jellies and foams are usually used along with copper releasing IUDs to increase their contraceptive value.

Statement B : Breakthrough bleeding and irregular menstrual bleeding are some of the ill-effects associated with the use of contraceptives.

Select the correct option

- (1) Only statement A is incorrect
- (2) Only statement B is incorrect
- (3) Both statements A and B are incorrect
- (4) Both statements A and B are correct

145. Choose the viral disease for which vaccine has been produced from yeast by RDT.

- (1) Hepatitis-B
- (2) Typhoid
- (3) Cancer
- (4) Rheumatoid arthritis

146. How many of the diseases mentioned in the box below require a vector for transmission of its causative pathogen?

Amoebiasis, Ascariasis, Ringworms, Diphtheria, Dengue, Malaria, Chikungunya, Filariasis

Choose the correct option.

- (1) Three
- (2) Four
- (3) Seven
- (4) Six

147. Match column I with column II w.r.t. number of disulfide bonds present in the monomeric structure of an antibody molecule.

	Column I		Column II
(a)	Within heavy chains	(i)	2
(b)	Within light chains	(ii)	6
(c)	Between heavy and light chains	(iii)	8
		(iv)	4

Select the **correct** option.

- (1) a(ii), b(iv), c(i)
- (2) a(iii), b(iv), c(i)
- (3) a(iv), b(iii), c(ii)
- (4) a(ii), b(iii), c(iv)

148. Select the odd one w.r.t. Australian marsupials.

- (1) Banded anteater
- (2) Bandicoot
- (3) Bobcat
- (4) Numbat

149. Read the following statements

- (a) According to WHO, reproductive health only refers to physical well-being.
- (b) Introduction of sex education in schools should be discouraged.
- (c) Indian laws permit legal adoption of orphan children.
- (d) Improved health facilities has promoted an explosive growth of population.

How many of the above statements is/are correct?

- (1) One
- (2) Two
- (3) Three
- (4) Four

150. The drugs obtained from all of the given plants have hallucinogenic properties at a particular dosage, **except**

- (1) *Erythroxylum coca*
- (2) *Datura*
- (3) *Atropa belladonna*
- (4) *Papaver somniferum*

151. Malignant tumor differs from benign tumor as

- (1) Former is less invading
- (2) Latter remains confined to the original location
- (3) Latter shows metastasis
- (4) Former shows contact inhibition

152. Select the extinct land reptile which had plates extending over the back and spikes at the end of the tail.

- (1) *Stegosaurus*
- (2) *Tyrannosaurus*
- (3) *Brachiosaurus*
- (4) *Pteranodon*

153. The antibodies present in colostrum are mainly of _____ type which provide _____ to the baby. Select the correct option to fill in the respective blanks.

- (1) IgE; Natural active acquired immunity
- (2) IgG; Artificial active acquired immunity
- (3) IgA; Natural passive acquired immunity
- (4) IgA; Artificial active acquired immunity

154. How many of the methods listed in the box below involve *in-vitro* fertilisation?

GIFT, AI, ZIFT, IUT, IUI

Choose the correct option.

- (1) Five
- (2) Two
- (3) Three
- (4) Four

155. Assertion (A) : Though all people are vulnerable to STIs, their incidences are reported to be high among persons in the age group of 15-24 years.

Reason (R) : Myths, lack of knowledge and misconceptions about sex related aspects, unsafe and unhygienic sexual practices contribute to venereal diseases.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

156. During post-industrialisation period in England

- (1) More white-winged moths survived due to thick growth of lichen on tree trunks
- (2) White-winged moths were eliminated from population completely
- (3) More dark-winged moths were able to survive due to thick growth of lichen on tree trunks
- (4) Dark-winged moth survived because tree trunks became dark and they could camouflage themselves better

157. Consider the following.

- (a) Site of production and differentiation of immature lymphocytes.
 - (b) Constitutes about 50 per cent of lymphoid tissue.
 - (c) Provides site for interaction of lymphocytes with antigen.
- Which of the above is/are not true for mucosa associated lymphoid tissue?

- (1) (a) only
- (2) (c) only
- (3) (a) and (c)
- (4) (a) and (b)

158. Which of the following is an additional benefit provided by hormone releasing IUDs over other types of IUDs?

- (1) Make cervix hostile to sperms
- (2) Suppress sperm motility
- (3) Increase phagocytosis of sperms within uterus
- (4) Reduce fertilising capacity of sperms

159. An American scientist conducted an experiment in which he created similar conditions in a closed flask in laboratory as were present on primitive Earth which led to the acceptance of theory of chemical evolution. This experiment was performed in the year

- (1) 1953
- (2) 1820
- (3) 1920
- (4) 1938

160. A little girl suffered from an infectious disease named 'X' for the first time and she took a week to recover from it. She was again infected with the pathogen of same disease after few months but she recovered within 4 days by this time. Which of the following options is incorrect w.r.t. the above given scenario?

- (1) Former immune response was of low intensity and slower but latter was of high intensity and rapid.
- (2) Former is called primary immune response and latter is called the anamnestic immune response.
- (3) Former is carried out with the help of B-lymphocytes and latter is carried out only with the help of T-lymphocytes.
- (4) Latter was produced due to the memory of first encounter.

161. Select the option representing incorrect feature w.r.t. emergency contraceptives.

- (1) Progestogen-estrogen combination can be administered
- (2) Progestogen only can be administered
- (3) They can inhibit ovulation and implantation
- (4) Should be administered after a week of unprotected intercourse for best effect.

162. Consider the following reproductive events in humans.

- (a) Gametogenesis
- (b) Gamete transport
- (c) Fertilisation
- (d) Gestation

Which of the above events will not occur in tubectomised female?

- (1) (a), (b), (c) and (d)
- (2) (a) and (d) only
- (3) (b), (c) and (d)
- (4) (a), (b) and (c)

163. Which of the following is not true for health?

- (1) Defined as state of complete physical, mental and social well being
- (2) Healthy people are more efficient at work
- (3) Health decreases longevity of people and reduces maternal and infant mortality
- (4) Health does not simply imply 'absence of disease'

164. The theory of evolution proposed by Ernst Haeckel was based upon the observation that

- (1) Embryos never pass through the adult stages of other animals
- (2) Comparative anatomy and morphology show similarities and differences among organisms
- (3) Certain features during embryonic stage common to all vertebrates that are absent in adult
- (4) The geological history of Earth closely correlates with the biological history of Earth

165. Forelimbs of whales and bats have similar anatomical structure, but perform different functions due to

- (1) Similar habitat
- (2) Convergent evolution
- (3) No common ancestor
- (4) Adaptations to different needs

166. Match column I with column II.

Column I	Column II
(a) <i>Dryopithecus</i>	(i) Brain capacity of 650-800 cc
(b) <i>Ramapithecus</i>	(ii) More ape-like
(c) <i>Homo erectus</i>	(iii) Probably ate meat
(d) <i>Homo habilis</i>	(iv) Hairy and walked like gorillas and chimpanzees

Choose the **correct** option.

- (1) a(ii), b(iv), c(i), d(iii)
- (2) a(iii), b(i), c(ii), d(iv)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(ii), b(iv), c(iii), d(i)

167. All of the given sets of organisms are causative agents of the same disease, **except**

- (1) *Streptococcus pneumoniae*, *Haemophilus influenzae*
- (2) *Microsporum*, *Epidermophyton*
- (3) *Ascaris*, *Entamoeba histolytica*
- (4) *W. malayi*, *W. bancrofti*

168. Read the following statements:

Statement A: About 2000 mya, the first cellular forms of life appeared on Earth.

Statement B: The first non-cellular forms of life could have originated 3 mya.

Select the **correct** option.

- (1) Both statements A and B are correct
- (2) Only statement B is correct
- (3) Only statement A is correct
- (4) Both statements A and B are incorrect

169. Complete the analogy by choosing the correct option w.r.t. direct ancestors of plants.

Conifer : Psilophyton :: Horsetails : _____

- (1) Rhynia type plant
- (2) Tracheophyte
- (3) Chlorophyte
- (4) Psilophyton

170. The couple should abstain from coitus from the day _____ of a 30 days menstrual cycle to prevent conception.

Select the correct option to fill the blank.

- (1) 12 to 19
- (2) 7 to 14
- (3) 16 to 23
- (4) 23 to 30

171. The theory which talks of a single huge explosion which attempts to explain to us the origin of Universe is

- (1) Spontaneous generation theory
- (2) Big Bang theory
- (3) Theory of special creation
- (4) Panspermia theory

172. Consider the following

- (a) Mucus coating of GIT
- (b) Saliva in mouth
- (c) Acid in stomach
- (d) Tears from eyes

Which of the above are included under physiological and physical barrier of innate immunity?

	Physiological barrier	Physical barrier
(1)	(a)	(b), (c), (d)
(2)	(a), (c)	(b), (d)
(3)	(b), (c), (d)	(a)
(4)	(b), (d)	(a), (c)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

173. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography is called

- (1) Saltation
- (2) Adaptive radiation
- (3) Analogy
- (4) Convergent evolution

174. Who among the following asserted that health is a state of body and mind where there was a balance of certain humors?

- (1) William Harvey
- (2) Charles Darwin
- (3) Ernst Heckel
- (4) Hippocrates

175. Select the incorrect statement w.r.t. the theory based on conventional religious literature.

- (1) All living species or types that we see today were created as such
- (2) The diversity was always the same since creation but will change in future
- (3) Earth is about 4000 years old
- (4) This theory was known as theory of special creation

176. Rocks form sediments and a cross-section of Earth's crust may indicate/contain

- (1) Fossils of same life-forms who never died during formation of particular sediment
- (2) Fossils of animals which did not appear at the time of sediment formation.
- (3) Arrangement of sediments one over the other during the long history of Earth
- (4) Fossils of different life forms which were present only in same aged rock sediment.

177. The use of condoms has been increased in recent years due to its additional benefit of

- (1) Protecting the user from contracting STDs
- (2) Zero chance of failure
- (3) Making uterus unsuitable for implantation
- (4) Making cervix hostile to the sperms

178. When we describe the story of life on Earth we treat evolution as a/an

- (1) Ongoing process occurring only due to saltation
- (2) Consequence of processes called natural selection
- (3) Unknown process due to only stabilising selection
- (4) Ongoing process due to asexual reproduction only

179. Dicotyledons and angiosperms are descendants of

- (1) Progymnosperms
- (2) Gnetales
- (3) Sphenopsids
- (4) Bryophytes

180. A monomeric antibody can be represented as

- (1) H₅L₅
- (2) H₂L₂
- (3) H₆L₄
- (4) H₄L₆